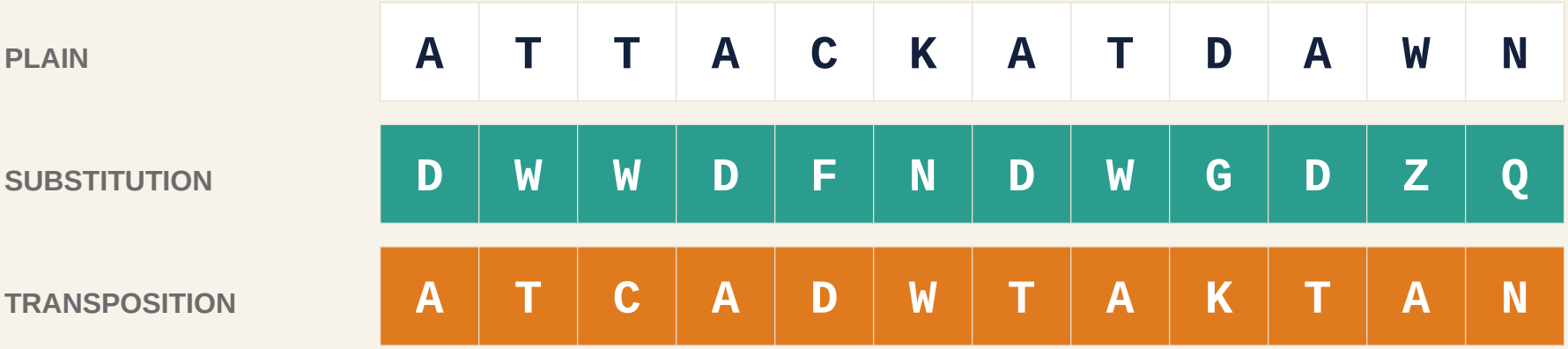
Substitution vs Transposition

Which Provides Better Security?

`new letters, same places` vs `same letters, new places`

A look at the two building blocks of classical (and modern) cryptography

Two Ways to Hide a Message



Substitution: Caesar shift of 3. Transposition: rail fence with 2 rails (odd letters, then even).

- Substitution changes what each letter is, but keeps the letters in order.
- Transposition keeps the same letters, but changes where they sit.
- Check: the letters of ATCADWTAKTAN are exactly the letters of ATTACKATDAWN.

Substitution: Strengths and Limits

Strengths

- Hides the identity of every letter.
- Huge key space: 26! (about 4 x 10^26) for a general alphabet.
- Polyalphabetic versions (Vigenère) flatten letter frequencies.
- With a random key as long as the message, it becomes the unbreakable one-time pad.

Limits

- A simple one-alphabet cipher keeps the language's frequency pattern, only relabelled.
- Letter order is untouched, so word lengths, doubled letters and common words show through.
- Repeating keys (Vigenère) leave a period that Kasiski and Friedman can detect.

Transposition: Strengths and Limits

Strengths

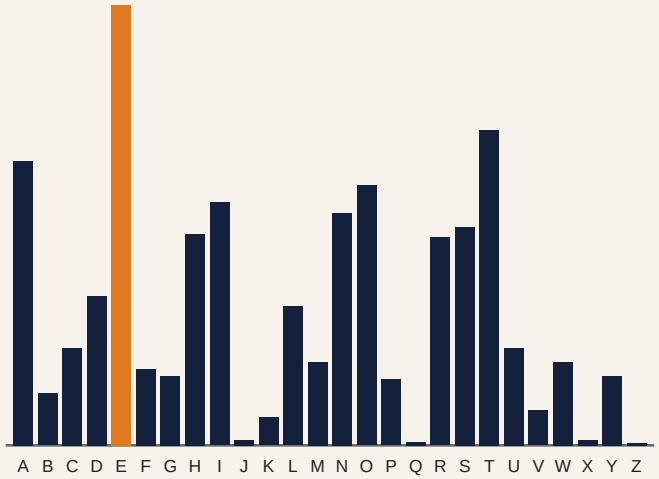
- Breaks up words, digraphs and phrases, so pairs like TH or QU vanish.
- No letter-substitution table to recover.
- A long, irregular rearrangement is hard to undo by hand.
- Doubling it (a second transposition) greatly raises the difficulty.

Limits

- Letter counts are unchanged, so the ciphertext frequencies match the plain language.
- That match is itself a clue that the cipher is a transposition.
- Attackers try anagramming, guess likely words, and test column counts.
- A short key gives few arrangements to try.

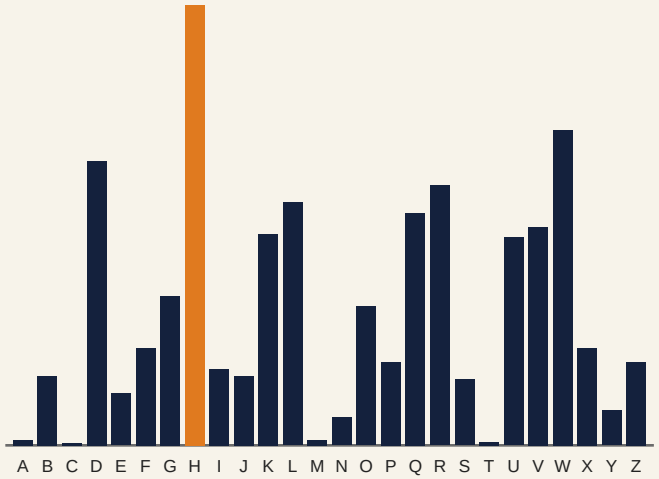
The Frequency Fingerprint

Plain English



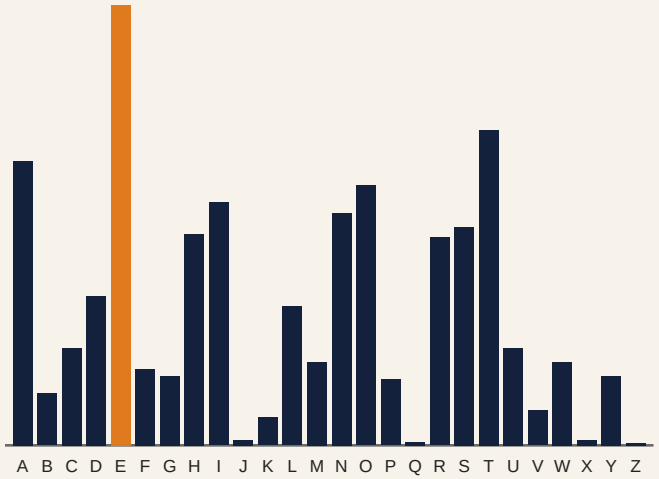
E is the tallest bar.

After Caesar shift (substitution)



Same shape, relabelled: the peak moves from E to H.

After transposition



Identical to plain English: nothing was relabelled.

Substitution disguises the letters. Transposition leaves the fingerprint untouched.

Key Space Compared

Cipher	Key	Number of keys
Caesar	A shift from 1 to 25	25 useful keys
Monoalphabetic substitution	Any rearrangement of the alphabet	26! (about 4 x 10^26)
Columnar transposition	An ordering of n columns	n! (n = 10: 3,628,800; n = 15: about 1.3 x 10^12)
Vigenère	A keyword of length m	26^m (m = 8: about 2.1 x 10^11)

A giant key space does not guarantee security: the monoalphabetic cipher falls to frequency analysis.

Shannon: Confusion and Diffusion

Confusion

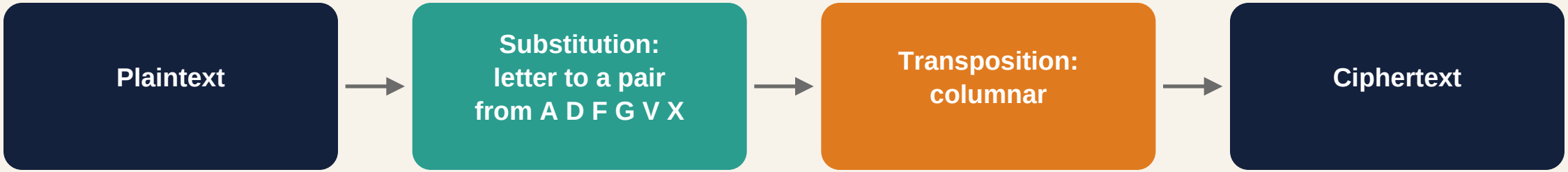
- Make the link between key and ciphertext complicated.
- Substitution is the classic tool.
- Alone, a simple substitution is still easy to relate back to the language.

Diffusion

- Spread the influence of each plaintext symbol across many ciphertext symbols.
- Transposition moves symbols around, which is a limited form of diffusion.
- Alone, it leaves every letter intact.

Claude Shannon (1949): strong ciphers combine both, and repeat them.

Combining Both: ADFGVX (1918)



- Introduced by Germany in 1918: a Polybius-square substitution followed by a columnar transposition.
- The substitution made letters into pairs of symbols; the transposition then scattered those pairs.
- It was broken in 1918 by the French cryptanalyst Georges Painvin, but it held out for a time.
- Lesson: a pairing of the two was far stronger than either part on its own.

Modern Ciphers Use Both

- DES uses 16 rounds of substitution and permutation steps; AES uses 10, 12 or 14 rounds.
- In AES, SubBytes is the substitution; ShiftRows and MixColumns spread the effect (diffusion).
- This is called a substitution-permutation network.
- Round after round, small changes in the plaintext or key change nearly everything in the output.

Scorecard

Property	Substitution	Transposition
What changes	The identity of letters	The position of letters
Letter frequencies	Disguised (relabelled, or flattened if polyalphabetic)	Unchanged: they match the language
Classic attack	Frequency analysis, Kasiski test	Anagramming, probable words, trying column counts
Shannon property	Confusion	Diffusion (limited)
Strong alone?	Only with many alphabets and a long or random key	No; better when doubled, still weak alone

So, Which Is Better?

- Taken alone, substitution is generally stronger: it changes the letters themselves, and polyalphabetic or random-key versions can be very strong.
- Transposition alone leaves letter frequencies exposed, so it is the weaker partner.
- Neither is enough by itself. Security comes from combining them and repeating them over many rounds.

Questions?